

Pre-School Setup Guide — Python & Anaconda

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Pre-School Setup Guide — Python & Anaconda

SAQUTI — Quantum Technologies School

Please complete this **at least a few days before the school starts**, ideally on the same laptop you'll bring with you. If anything fails, get in touch with the organisers *before* you arrive — troubleshooting installs during the session eats time that should go to the actual content.

Time required: 20–40 minutes, depending on your internet connection (the Anaconda installer is ~700 MB–1 GB).

What you're installing, and why

- ♦ **Anaconda** — a Python distribution that bundles Python, `conda` (an environment/package manager), Jupyter, and most of the scientific packages we'll use (NumPy, SciPy, Matplotlib), all in one installer.
- ♦ A dedicated **conda environment** called `saqutischool`, pinned to **Python 3.14** — an isolated sandbox for this school's packages, so it can't conflict with anything else already on your machine.
- ♦ **Qiskit** (and `qiskit-aer`, the local simulator) — installed with `pip` inside that environment, for later sessions.

Version policy: we pin the environment to **Python 3.14**, but deliberately do **not** pin package versions — every `conda install` / `pip install` command below installs the latest release available for Python 3.14 at the time you run it. If you set this up weeks before the school and a package has since released a new version, just re-run the same install command to pick it up (see the “Keeping things up to date” note at the end of this guide).

You do **not** need to already know how to code, or understand what any of this means yet — session 1 covers that. This guide only gets your machine ready.

Step 1 — Install Anaconda

Download the **Anaconda Distribution** installer for your operating system from anaconda.com/download (a free account/email may be requested — that's fine, no purchase involved). Choose the **64-bit graphical installer** for Python 3.x (not Miniconda, unless you're already comfortable managing packages manually).

Windows

1. Run the downloaded `.exe` installer.
2. Choose **“Just Me”** when asked (avoids needing admin rights).
3. Leave the install location at its default.
4. On the **Advanced Installation Options** screen, you can leave “Add Anaconda to my PATH environment variable” **unchecked** (Anaconda's own installer recommends this) — you'll instead use the **Anaconda Prompt**, which is added to your Start Menu automatically and doesn't need PATH changes.
5. Finish the install. Open **Anaconda Prompt** from the Start Menu (search “Anaconda Prompt”) for every command below — not the regular Command Prompt or PowerShell.

macOS

1. Run the downloaded `.pkg` installer and follow the prompts (this installs into your home directory by default — no admin password should be required, though macOS may still ask

you to confirm).

2. Open **Terminal** (Applications → Utilities → Terminal, or search with Spotlight).
3. Confirm the install by running:

```
conda --version
```

If this prints something like `conda 24.x.x`, you're set. If you get `command not found`, close and reopen Terminal (the installer updates your shell profile, which only takes effect in new terminal windows), then try again.

Linux

1. Run the downloaded `.sh` installer from a terminal:

```
bash Anaconda3-2024.xx-Linux-x86_64.sh
```

(substitute the actual filename you downloaded).

2. Press Enter/Space through the license, type `yes` to accept it, accept the default install location, and answer `yes` when asked whether the installer should run `conda init`.
3. Close and reopen your terminal, then confirm:

```
conda --version
```

Step 2 — Create the school's environment

From **Anaconda Prompt** (Windows) or **Terminal** (macOS/Linux), run:

```
conda create -n saqutischool python=3.14
```

Type `y` and press Enter when prompted to proceed. This downloads and sets up an isolated Python 3.14 environment — it won't touch anything else on your system.

Activate it:

```
conda activate saqutischool
```

Your prompt should now show `(saqutischool)` at the start of the line. **You'll need to run this `conda activate saqutischool` command every time you open a new terminal window for this school** — it doesn't stay active permanently.

Step 3 — Install the packages we'll use

With the environment active (you see `(saqutischool)` in your prompt):

```
conda install numpy scipy matplotlib jupyter jupyterlab -y
```

This will take a few minutes. Once it finishes, install Qiskit with `pip` (used for packages not distributed via conda):

```
pip install qiskit qiskit-aer pylatexenc
```

(`pylatexenc` lets Qiskit draw circuits as Matplotlib figures, which the notebooks use.)

Step 4 — Verify everything works

Still inside the activated `saqutischool` environment, launch Jupyter:

```
jupyter notebook
```

(or `jupyter lab` if you prefer the JupyterLab interface — either is fine for this school). This should open a browser tab showing a file browser. If your browser doesn't open automatically, copy the `http://localhost:8888/...` URL printed in the terminal into your browser manually.

Create a new Python 3 notebook (**New → Python 3 (ipykernel)**) and run the following in a single cell (**Shift+Enter** to run it):

```
import sys, numpy, scipy, matplotlib, qiskit
print("Python:", sys.version)
print("NumPy:", numpy.__version__)
print("SciPy:", scipy.__version__)
print("Matplotlib:", matplotlib.__version__)
print("Qiskit:", qiskit.__version__)

import matplotlib.pyplot as plt
plt.plot([0, 1, 2, 3], [0, 1, 4, 9])
plt.title("If you can see this plot, you're ready for SAQUTI!")
plt.show()
```

Success looks like: the Python line starts with `3.14`, four package version numbers print without errors, and a small chart appears below the cell. If you see that, you're fully set up — close Jupyter (Ctrl+C in the terminal, or just close the browser tab and terminal) and you're done.

Troubleshooting

Problem	Likely fix
<code>conda</code> : command not found (macOS/Linux)	Close and reopen your terminal so the installer's shell changes take effect. Still failing? Re-run the Anaconda installer and confirm it finished without errors.
<code>conda</code> not recognised (Windows)	Make sure you're using Anaconda Prompt from the Start Menu, not regular Command Prompt/PowerShell.
<code>conda create -n saqutischool python=3.14</code> fails to find a Python 3.14 build	Your Anaconda install may bundle an older <code>conda/conda-forge</code> index that hasn't caught up yet. Run <code>conda update -n base conda</code> first, then retry. Still stuck? Add the conda-forge channel explicitly: <code>conda create -n saqutischool -c conda-forge python=3.14</code> .
<code>pip install qiskit</code> fails / times out	Check you're connected to the internet and not on a restrictive network (some corporate/university Wi-Fi blocks package downloads) — try a personal hotspot if on campus Wi-Fi fails.

Problem	Likely fix
Jupyter opens but the kernel says “no module named qiskit” (or numpy/scipy)	You likely installed packages in a different environment than the one the notebook is using. In Jupyter, go to Kernel → Change Kernel and pick the one matching <code>saqutischool</code> (or restart with <code>saqutischool</code> activated in the terminal before running <code>jupyter notebook</code>).
Installer needs admin/sudo password you don’t have	Choose the “Just Me” / user-only install option rather than “All Users” — this avoids needing admin rights entirely.
Very slow / limited internet at home	The Anaconda installer and package downloads total roughly 1.5–2 GB. Do this on the best connection available to you, well before travelling to the school.

If you’re still stuck after trying the above, **email the organisers before the school** with: 1. Which operating system and version you’re on. 2. Which step you’re stuck on. 3. The exact error message (a screenshot is great).

Keeping things up to date

Because none of the install commands above pin a package version, running them again always pulls the current latest release for Python 3.14. If you set your environment up well ahead of the school and want to refresh it right before you travel:

```
conda activate saqutischool
conda update numpy scipy matplotlib jupyter jupyterlab -y
pip install --upgrade qiskit qiskit-aer pylatexenc
```

Quick reference — commands you’ll use again

```
conda activate saqutischool    # run this every time you start work for the school
jupyter notebook              # or: jupyter lab
```

To leave the environment when you’re done:

```
conda deactivate
```

See you at SAQUTI!